



GITHUB PRESENCE & FIRST CONTRIBUTION

Build, Contribute, and Solve

TRACK

Backend .NET

PARTS

3 — Calculator, PR,
LeetCode

WEIGHT

Graded Assignment

Objective: Publish a working project on your own GitHub profile, contribute to a shared repository the proper way — fork, SSH, branch, and pull request — and sharpen your problem-solving skills on LeetCode using C#.

01 Calculator Project

Create a public GitHub repository named `calculator` and build a Calculator application using any language or technology you prefer — for example a C#, Java, or Python console application, or an HTML/CSS/JavaScript calculator.

Required Features

- Take two numbers from the user.
- Support addition, subtraction, multiplication, and division.
- Display the result clearly.
- Handle division by zero and invalid input reasonably.
- Allow the user to perform another calculation without restarting the application.

Git Requirements

Do not build the entire project directly on `main`. Use branches with common naming conventions such as `feature/basic-calculator`, `feature/input-validation`, and `docs/update-readme`, then merge your completed work into `main`. Your repository must include at least 3 meaningful commits, e.g. `feat: initialize calculator project`, `fix: handle division by zero`, `docs: add project README`.

Repository Requirements

- Clean, working source code and a suitable `.gitignore`.
- A `README.md` with project description, language/technologies used, how to run it, and a screenshot or GIF showing it working.
- No secrets, API keys, `.env` files, build output, or dependency folders (`node_modules`, `bin` / `obj`).

GitHub Profile Checklist

- ✓ Clear profile photo or avatar
- ✓ Your name and a short professional bio
- ✓ LinkedIn or portfolio link, if available

02 Fork, SSH & Website Contribution

Contribute a small website to the class repository: <https://github.com/SimulationEG/github-contributors>

Workflow

- Fork the class repository to your own GitHub account.
- Add an SSH key to your GitHub account and clone your fork using SSH.
- Do not push directly to the original repository.
- Create a branch named `feature/add-your-github-username` (e.g. `feature/add-ahmed-mohamed`).
- Create your own folder at `students/your-github-username` and build a small website inside it — HTML only, HTML + CSS, or HTML + CSS + JavaScript.

Website Idea

Choose any simple idea: a personal developer profile, mini portfolio, About Me page, study journey page, technology roadmap, interactive greeting page, simple quiz, quote generator, to-do list, or mini calculator.

Website Content

- Your full name, GitHub username, and learning track.
- A link to your Calculator repository and your GitHub profile.
- A LinkedIn link, if available, and a clean, readable layout.

```
students/your-github-username/  
  index.html  
  style.css  
  script.js
```

Add your name and website link to the main contributor list in the starter repository, then push your branch and open a Pull Request from `YOUR-USERNAME:feature/add-your-github-username` into `SimulationEG:main`, titled `feat: add [Your Full Name] contributor website`.

03 LeetCode Problem Solving

Solve **Reverse String** using C#.

- Use the Two Pointers approach.
- Do not use built-in methods that reverse the array.
- Modify the input `char[]` in place.
- Add a screenshot showing the submission is Accepted.
- Include a short explanation of the solution idea, time complexity, and space complexity.

```
Expected Time Complexity: O(n)  
Expected Space Complexity: O(1)
```